Python Activity Record

Nested Loop Calendar Generator

1. Problem Statement

Create a Python program that accepts a month name, the number of days in that month, and its starting weekday. The program should validate the number of days (28–31) and weekday index (Monday = 0 to Sunday = 6), then display a calendar layout with weekdays and dates. If the values are outside the allowed ranges, it should print an error message.

2. Objectives

- To accept month details from the user.
- To validate the month length and starting weekday.
- To use a loop to display dates from 1 through the given number of days.
- To arrange dates into weekday rows using spacing and a modulo condition.
- To display an error message for invalid values.

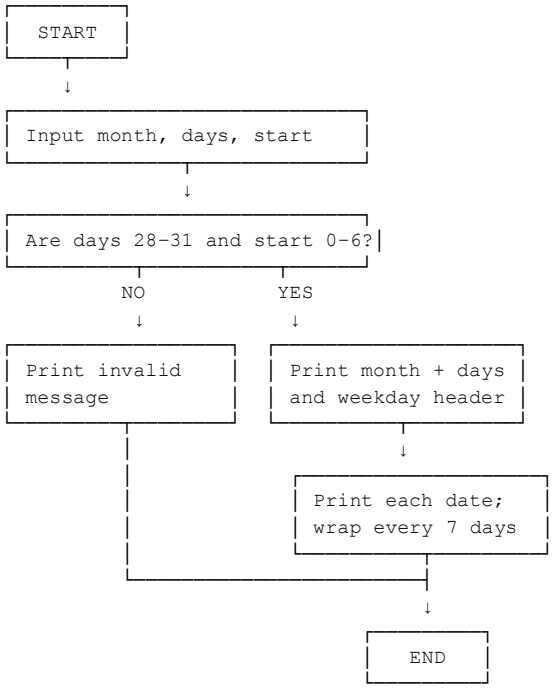
3. Assumptions

- The user enters a month name as text.
- The number of days is entered as an integer between 28 and 31.
- The starting weekday is an integer from 0 to 6, where Monday is 0 and Sunday is 6.
- The program does not verify whether the chosen month actually has the supplied number of days or whether the starting weekday matches a real calendar.
- Non-integer values for days or starting weekday are not handled and may raise a ValueError.

4. Algorithm

1. Start.
2. Input the month name.
3. Input the number of days in the month.
4. Input the starting weekday index.
5. If days is below 28 or above 31, or start is below 0 or above 6, print the invalid-input message.
6. Otherwise, print the month name and weekday headings.
7. Print initial spaces based on the starting weekday.
8. Loop through each date from 1 to days and print the date.
9. When (start + date) modulo 7 equals 0, move to a new line.
10. Print a final newline and stop.

5. Flowchart



6. Pseudocode

```
BEGIN
    INPUT month
    INPUT days
    INPUT start

    IF days < 28 OR days > 31 OR start < 0 OR start > 6 THEN
        PRINT "Invalid month length or weekday."
    ELSE
        PRINT month
        PRINT "Mo Tu We Th Fr Sa Su"
        PRINT "   " repeated start times, without newline

        FOR date = 1 TO days
            PRINT date formatted in 2 spaces, without newline
            PRINT one space, without newline
            IF (start + date) MOD 7 == 0 THEN
                PRINT newline
            END IF
        END FOR
        PRINT newline
    END IF
END
```

7. Python Source Code

```
month = input("Month name: ")
```

```
days = int(input("Days in month (28-31): "))

start = int(input("Starting weekday (Mon=0, Sun=6): "))

if days < 28 or days > 31 or start < 0 or start > 6:

    print("Invalid month length or weekday.")

else:

    print(month)

    print("Mo Tu We Th Fr Sa Su")

    print("   " * start, end="")

    for date in range(1, days + 1):

        print(f"{date:2}", end=" ")

        if (start + date) % 7 == 0:

            print()

    print()
```

8. Test Cases

Test	Input	Expected Output	Purpose
1	October, 31, 3	Calendar dates printed in weekday rows	Valid input
2	February, 32, 0	Invalid month length or weekday.	Days above maximum
3	May, 30, 7	Invalid month length or weekday.	Weekday index above maximum
4	June, 28, 0	Calendar with 28 dates	Minimum valid days
5	April, 30, -1	Invalid month length or weekday.	Negative weekday index

9. Output Screenshots

Screenshot 1: Valid calendar input.

```
OUTPUT SCREENSHOT 1 — VALID INPUT
------------------------------------------
$ python "Nested Loop Calendar Generator.py"
Month name: October
Days in month (28-31): 31
Starting weekday (Mon=0, Sun=6): 3

October
Mo Tu We Th Fr Sa Su
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

Screenshot 2: Invalid number of days.

```
OUTPUT SCREENSHOT 2 — INVALID INPUT
------------------------------------------
$ python "Nested Loop Calendar Generator.py"
Month name: February
Days in month (28-31): 32
Starting weekday (Mon=0, Sun=6): 0

Invalid month length or weekday.
```

10. Conclusion

The Nested Loop Calendar Generator demonstrates user input, validation using conditional statements, loops, formatted printing, and modulo arithmetic. It prints a simple calendar-style layout for accepted inputs and displays an error message for invalid day counts or weekday indexes.

Limitation: The program checks allowed ranges only. It does not verify the actual number of days for the named month or calculate the weekday from a real date.